

Charotar University of Science & Technology (CHARUSAT)
Third Semester of B.Tech.(IT) Theory Examination (March-April 2018)
MA 207 Discrete Mathematics and Applications

Date: 26/03/2018, Monday **Time:** 01:30 p.m. to 04:30 p.m **Maximum Marks:** 70

INSTRUCTIONS:

1. The question paper comprises two sections.
 2. Section I and II must be attempted in separate answer sheets.
 3. Make suitable assumptions and draw neat figures wherever required.
 4. Use of scientific calculator is allowed.
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SECTION-I

Q.1. Choose the correct answer for the following questions. [07]

1. $p \Rightarrow q \equiv$ _____.

- ☐ (A) $\sim p \wedge q$ ☐ (B) $\sim p \vee q$ ☐ (C) $p \vee q$ ☐ (D) $p \wedge q$.

2. The rule $p \Rightarrow (p \wedge q)$ is called _____.

- ☐ (A) addition rule ☐ (C) Conjunction rule
☐ (B) simplification rule ☐ (D) modus ponens.

3. Every distributive lattice is also _____ lattice.

- ☐ (A) modular ☐ (C) complete
☐ (B) complemented ☐ (D) bounded

4. Which of the following is true for Boolean algebra?

- ☐ (A) Boolean algebra is a bounded lattice
☐ (B) Boolean algebra is not a distributive lattice
☐ (C) Boolean algebra is a modular lattice
☐ (D) Boolean algebra is not a complemented lattice

5. Which of the following properties is not true for partially ordered set?

- ☐ (A) reflexivity ☐ (C) antisymmetry
☐ (B) symmetry ☐ (D) transitivity

6. Let $(L, *, \oplus)$ be any lattices and $a \in L$. Then $a \oplus a =$ _____.

Ⓐ 0

Ⓑ 1

Ⓒ a

Ⓓ a^2 .

7. Which of the following is symmetric Boolean expression?

Ⓐ $(a * b') \oplus b$

Ⓒ $(a * b') \oplus (a' * b)$

Ⓑ $(a * b') \oplus b'$

Ⓓ $(a * b') \oplus (a \oplus b')$.

Q.2 Attempt any four questions.

[16]

- (i) Prove that s is a valid conclusion from the premises $p \Rightarrow q, p \Rightarrow r, \sim (q \wedge r)$ and $s \vee p$.
- (ii) Prove that $\{\vee, \sim\}$ is a minimal functionally complete set of connectives.
- (iii) Define Boolean algebra. Show that there is no Boolean algebra with three elements.
- (iv) Define join irreducible elements and meet irreducible elements. Find out Join irreducible elements and meet irreducible elements from following figure.

(v) State and prove associative property for Lattice.

(vii) State and prove Isotonicity property for Lattice.

Q.3(a) Attempt any two questions.

[06]

- (ii) Obtain the principle conjunctive normal form of $(\sim p \Rightarrow r) \wedge (q \Leftrightarrow p)$ without using truth table.
- (ii) State and prove Modular inequality for lattice.
- (iii) Consider a Boolean algebra $B = S_{10}$. Define a Boolean function $f : B^3 \rightarrow B$ as $f(x, y, z) = x'yz' + xyz + x'yz' + x'yz$. Find image of $(1, 2, 5)$ for this function.

(b) Attempt any one of the following questions.

[06]

(i) Let B be a Boolean algebra and $a, b \in B$. Then prove that

$$(i)(a + b)' = a'b' \quad (ii)(ab)' = a' + b'.$$

(ii) Define an equivalence relation. If R be relation in the set of integers $\mathbb{Z}$ defined by $R = \{(x, y) : x \in \mathbb{Z}, y \in \mathbb{Z}, (x - y) \text{ is divisible by } 3\}$. Then prove that R is an equivalence relation.

SECTION-II

Q.4 Choose the correct answer for the following questions.

[07]

1. A nonempty connected graph is Eulerian if and only if its vertices are all _____.

(A) odd degree

(C) zero degree

(B) even degree

(D) none of these.

2. Euler's formula is given by _____.

(A) $n - e + r = 2$

(C) $e - n + r = 2$

(B) $n - e - r = 2$

(D) none of these.

3. If $G = (V, E)$ be an undirected graph with e -edges, then $\sum_{v \in V} \deg_G(v)$ is _____.

(A) twice total no. of edges

(C) less than total no. of edges

(B) equal total no. of edges

(D) more than total no. of edges

4. Complete graph with n vertices has _____ edges.

(A) $\binom{n}{0}$

(B) $\binom{n}{1}$

(C) $\binom{n}{2}$

(D) $\binom{n}{n}$

5. $K_{3,3}$ is _____.

(A) planar graph

(C) Hamiltonian graph

(B) non planar graph

(D) Eulerian graph

6. Tree with n vertices and _____ edges.

(A) n

(B) $n + 1$

(C) $n - 1$

(D) $n - 2$

7. Full binary tree in which each internal vertex has _____.

Ⓐ exactly one child

Ⓒ no child

Ⓑ exactly 4 children

Ⓓ exactly two children.

Q.5 Attempt any four questions.

[16]

(i) Check whether the following graphs are isomorphic or not.

(ii) C_6 is bipartite graph or not. Justify your answer.

(iii) Define Region of graph. A connected planar graph has 10 vertices each of degree 3. Into how many regions, does a representation of this planar graph split the plane?

(iv) Find all spanning tree for the graph G shown in figure by removing the edges in simple circuit.

(v) Find minimal spanning tree of following graph using Kruskal's algorithm.

- (vi) Solve $a_n = a_{n-1} + 2a_{n-2}, n \geq 2$ with initial conditions $a_0 = 0$ and $a_1 = 1$.

Q.6(a) Attempt any two questions.

[06]

- (i) Define bipartite graph, Isomorphic graph, minimal spanning tree.
- (ii) What are the left and right children of b shown in figure? what are left and right subtrees of a ?

- (iii) Represent binary tree of expression

$$A * B - C \uparrow D + E / F.$$

(b) Attempt any one question.

[06]

- (i) Use generating function to solve recurrence relation $a_n = 3a_{n-1} + 2, a_0 = 1$.
- (ii) Use Dijkstra's algorithm to find shortest path from A to other vertices of the following graph.

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